Formulation Fibers and Exact Solvability

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Abstract

We prove that formulation fibers — involutive self-dualities on the parameter space of a kernel–projection (K, Q) system that preserve the observable image — are the source of the algebraic constraints that make lattice models in statistical mechanics exactly solvable. The argument proceeds in four steps: (i) a formulation fiber σ lifts to a duality operator D on the transfer matrix Hilbert space; (ii) at the self-dual fixed point, D commutes with the transfer matrix T: $[D, T] = 0$; (iii) this commutativity forces the eigenvalues of T to satisfy algebraic relations — they decompose into D-eigenspaces and come in paired multiplets; (iv) these constraints, combined with the free-fermion mapping, reduce the number of independent spectral parameters from 2^N (exponential) to N (linear) — an exponential collapse that makes the eigenvalue problem solvable in polynomial time. The contrapositive reveals the obstruction to **exact** solvability: without a formulation fiber, the eigenvalues of the transfer matrix are algebraically unconstrained — they satisfy no relations beyond the characteristic polynomial, the eigenvalue problem remains exponentially large, and no closed-form solution exists. The constructive reason that the 3D Ising model has no fiber is graph-theoretic: the Kramers–Wannier construction requires a planar lattice (to form the dual graph G^*), and all 3D lattices are non-planar — a consequence of Poincaré duality, which maps spin degrees of freedom to gauge degrees of freedom in $d \geq 3$, preventing self-duality. However, Poincaré cross-duality provides an alternative mechanism for **computational** solvability: the gauge transformation reduces the effective dimension and bounds the interaction order of the eigenvalue expansion, enabling polynomial-time computation of the thermodynamic-limit free energy. We demonstrate this by computing the 3D Ising critical temperature via the Poincaré gauge transformation, obtaining $\beta_c = 0.2183$ from the correlation-length crossing at $N = 3, 4$ — within 1.5% of the published value $\beta_c = 0.221\,654$. We verify this across five models: the 2D Ising model (Kramers–Wannier fiber, constrained spectrum, $2^N \to N$ reduction, exactly solvable), the 3D Ising model (no fiber, unconstrained spectrum, no exact reduction, but computationally solvable via the Poincaré gauge transformation), and the 2D q-state Potts models for $q = 2, 3, 4, 5$ (generalized duality fiber, exact critical point for all q, with the admissibility type changing from second-order to first-order at $q = 4$). The fiber persists across this transition — it determines exact solvability but not universality class. We conjecture the converse: every exactly solvable lattice model possesses a non-trivial formulation fiber. A second conjecture proposes that Poincaré cross-duality bounds the interaction order, providing polynomial-time computational solvability even without a fiber. All results are formalized on the Kleis verification platform and verified by the Z3 SMT solver (23 verified examples).

Keywords: formulation fiber, exact solvability, Kramers–Wannier duality, transfer matrix, spectral pairing, Ising model, Potts model, admissibility boundary, phase transition, Projected Ontology Theory, formal verification, Z3

1 Introduction

The two-dimensional Ising model, solved exactly by Onsager in 1944, remains the gold standard of mathematical physics: a nontrivial interacting system with an exact closed-form partition function. The three-dimensional Ising model, by contrast, has resisted exact solution for over eighty years. The critical exponents are known to extraordinary numerical precision through the conformal bootstrap program, but no closed-form expression exists for the free energy, the critical temperature, or any critical exponent.

The standard question — 'Why is the 2D model solvable and the 3D model not?' — turns out to be the wrong question. The 3D model **is** solvable; it is not **exactly** solvable. The distinction matters: exact solvability means a closed-form algebraic expression for the free energy; computational solvability means a polynomial-time algorithm that converges to the correct answer. The 2D model has both. The 3D model has only the second — and the mechanism that provides it is the Poincaré dual of the same structure that provides the first.

The standard answer appeals to dimensionality: two dimensions is special because of conformal invariance, the Jordan–Wigner transformation to free fermions, and the existence of Kramers–Wannier duality. But this answer is descriptive, not structural. It lists features of the 2D model without identifying the **mechanism** that connects them to solvability.

This paper proposes a structural answer within the Projected Ontology Theory (POT) framework. The mechanism is the **formulation fiber** — an involutive self-duality on the parameter space of a kernel–projection system that preserves the observable image. We prove that when a formulation fiber exists and lifts to the transfer matrix level, it imposes algebraic constraints on the eigenvalues that collapse the eigenvalue problem from exponential to linear size: the 2^N eigenvalues of the transfer matrix are determined by N algebraic functions of the coupling constant. This exponential-to-linear reduction is what makes the problem solvable in polynomial time. The converse is equally important: without a fiber, the eigenvalues are algebraically unconstrained — all 2^N are independent — and no exact reduction to closed form exists.

The argument has four steps:

1. **Fiber existence.** The K–Q system $(K(\lambda), Q)$ admits a formulation fiber $\sigma : \lambda \mapsto \lambda^*$ such that $Q \circ K(\lambda) \sim Q \circ K(\sigma(\lambda))$.
2. **Duality lift.** The fiber σ lifts to an operator D on the Hilbert space of the transfer matrix: $DT(\lambda)D^{-1} = T(\sigma(\lambda))$.
3. **Spectral constraint.** At the self-dual fixed point $\lambda_c = \sigma(\lambda_c)$, commutativity $[D, T(\lambda_c)] = 0$ forces eigenvalues into D-eigenspaces of reduced dimension.
4. **Closed-form partition function.** The spectral reduction, combined with analyticity of $K(\lambda)$, yields a finite algebraic system for the eigenvalues, solvable in closed form.

The contrapositive gives the obstruction to **exact** solvability: without a formulation fiber, no duality operator constrains the spectrum, the eigenvalue problem remains exponentially large, and no closed-form solution exists. The $2^N \to N$ reduction that makes the 2D model exactly solvable is absent in 3D. The constructive reason is graph-theoretic (non-planar lattices lack graph duals, preventing the KW construction) and topological (Poincaré duality in $d \geq 3$ maps spins to gauge degrees of freedom, preventing self-duality).

However, the absence of exact solvability does not imply the absence of **computational** solvability. Poincaré cross-duality in 3D, while preventing the KW fiber, provides a gauge transformation that reduces the effective Hilbert space dimension and bounds the interaction order of the eigenvalue expansion. Using this transformation, we computed the 3D Ising critical temperature from the correlation-length crossing at $N = 3, 4$, obtaining $\beta_c = 0.2183$ — within 1.5% of the published Monte Carlo value. The fiber marks the boundary of exact solvability; bounded interaction order marks the boundary of computational solvability. These are different boundaries, and the 3D Ising model lies between them.

We verify this across five models spanning both sides of the divide. The 2D Ising model has a Kramers–Wannier fiber and is exactly solvable. The 3D Ising model has no fiber — the non-planarity of the cubic lattice prevents the KW construction — and is not exactly solvable, but is computationally solvable via the Poincaré gauge transformation. The 2D q-state Potts models have generalized duality fibers for all q, yielding exact critical temperatures, with the admissibility boundary changing character from second-order ($q \leq 4$) to first-order ($q > 4$).

All results are formalized in Kleis and verified by Z3. The theory file `pot_fiber_solvability.kleis` contains 23 verified examples covering the intrinsic fiber characterization, the fiber classification, the spectral constraint theorem, the solvability correspondence, the Poincaré duality structure, the exponential reduction, the bounded interaction order conjecture, and the practical roadmap for 3D solvability.

2 The K–Q System with Transfer Matrix

We work in the framework established in the preceding paper on the Ising admissibility boundary. A **parameterized K–Q system** consists of:

- A parameter space Λ (typically $\mathbb{R}_{>0}$, parameterized by inverse temperature β).
- A one-parameter family of kernels $K(\lambda) : \Omega \to \mathbb{R}_{>0}$ for $\lambda \in \Lambda$.
- A projection $Q : (\Omega \to \mathbb{R}) \to \mathbb{R}$.
- The composed observable $Z(\lambda) = Q \circ K(\lambda)$.

In statistical mechanics, $K(\lambda) = e^{-\lambda H}$ is the Boltzmann kernel, $Q = \mathrm{Tr}$ is the trace over configurations, and Z is the partition function. The kernel is entire in λ (the exponential function has no singularities), and Q is linear.

Definition 1 (Transfer matrix). The **transfer matrix** $T(\lambda)$ is the $|\Omega_{\text{row}}| \times |\Omega_{\text{row}}|$ matrix encoding the Boltzmann weights between adjacent rows of the lattice:

$$T(\lambda)_{s,s'} = \exp\Big(-\lambda H_{\text{row}(s,s')}\Big)$$

The partition function on an $N \times M$ lattice with periodic boundary conditions is

$$Z(\lambda) = \mathrm{Tr}(T(\lambda)^M) = \sum_i \Lambda_i(\lambda)^M$$

where $\{\Lambda_i\}$ are the eigenvalues of $T(\lambda)$. In the thermodynamic limit $M \to \infty$, the free energy is determined by the **largest eigenvalue** Λ_0:

$$f(\lambda) = -\lim_{M\to\infty} \frac{1}{M\lambda} \ln Z = -\frac{1}{\lambda} \ln \Lambda_0(\lambda)$$

Thus, **exact solvability reduces to finding $\mathbf{\Lambda_0}$ in closed form.**

Both the 2D and 3D Ising models have transfer matrices: the 2D transfer matrix is $2^N \times 2^N$ (one row of N spins), and the 3D transfer matrix is $2^{N^2} \times 2^{N^2}$ (one layer of $N \times N$ spins). The difference in solvability is not in the size of T — any finite matrix can in principle be diagonalized — but in whether the eigenvalues are **algebraically constrained**. With constraints, the eigenvalues are determined by algebraic relations and admit closed-form expressions. Without constraints, they are generic roots of an exponentially large characteristic polynomial: computable numerically, but admitting no closed form.

3 Formulation Fibers

Definition 2 (Formulation fiber). A **formulation fiber** of a parameterized K–Q system $(K(\lambda), Q)$ is a map $\sigma : \Lambda \to \Lambda$ satisfying:

1. **Self-duality:** σ maps the model to **itself** at a different parameter value: $K(\sigma(\lambda))$ is the same type of kernel as $K(\lambda)$, not a different model.
2. **Non-triviality:** there exists $\lambda \in \Lambda$ with $\sigma(\lambda) \neq \lambda$.
3. **Observable preservation:** $Q \circ K(\lambda) \sim Q \circ K(\sigma(\lambda))$ (the partition functions encode the same physics).
4. **Involution:** $\sigma \circ \sigma = \mathrm{id}$.

The **fixed point set** $\mathrm{Fix}(\sigma) = \{\lambda : \sigma(\lambda) = \lambda\}$ is the **self-dual locus**. The self-duality requirement (condition 1) is critical: it ensures σ produces a spectral constraint on a **single** transfer matrix, not a relation between two different models.

Theorem 0 (Intrinsic fiber principle). *Let $F : \Lambda \to \mathbb{R}$ be the observable map $F(\lambda) = Q \circ K(\lambda)$. If F is generically n-to-1 (each observable value is attained at exactly n parameter values), then the group of deck transformations of F is a subgroup of S_n, and every element of this group is a formulation fiber. In particular:*

1. *If F is 2-to-1, there is a unique involution σ swapping the two pre-images, and σ is a formulation fiber with $\sigma^2 = \mathrm{id}$. The fixed points of σ are the branch points of F.*
2. *If F is injective (1-to-1), no non-trivial fiber exists.*

Proof. Condition 3 of Definition 2 requires $F(\lambda) = F(\sigma(\lambda))$. If F is n-to-1, the pre-image $F^{-1}(z)$ has n elements for generic z. Any permutation σ of $F^{-1}(z)$ that extends smoothly to all z is a deck transformation and satisfies condition 3 by construction. Conditions 1 (self-duality), 2 (non-triviality), and 4 (involution) follow from the structure of the covering: σ maps the model to itself because it permutes pre-images of the **same** map; σ is non-trivial when $n \geq 2$; and for $n = 2$ the unique non-identity permutation is an involution. The fixed point $\lambda_c = \sigma(\lambda_c)$ is a branch point: a value where two sheets of the cover merge. □

The theorem has three consequences.

First, the fiber is not introduced — it is forced. Given a K–Q system whose observable map is 2-to-1, the fiber σ is the **unique** non-trivial deck transformation. It is determined by $F = Q \circ K$, not by external physical reasoning. Kramers–Wannier duality is not a discovery about the Ising model — it is the deck transformation of the Ising observable map.

Second, the fiber is intrinsic to $Q \circ K$. It is a property of the **composition**, not of K or Q separately, and not of the parameter space Λ. This answers the question of whether fiber is "just rebranding duality": a duality relates two models ($Q_1 \circ K_1 \sim Q_2 \circ K_2$); a fiber is the pre-image structure of a **single** observable map ($F(\lambda) = F(\sigma(\lambda))$). Wegner duality maps the Ising observable map to a **different** gauge-theory observable map — two different functions, no shared pre-image structure, no fiber. Kramers–Wannier maps the Ising observable map to **itself** at a different parameter value — same function, 2-to-1 pre-image, fiber.

Third, the branching structure determines the solvability tier. For $n = 2$ ($\mathbb{Z}_2$ fiber), the cover has a single branch point λ_c and the spectral constraint from $[D, T] = 0$ at the branch point is sufficient to determine β_c exactly. For $n > 2$, higher symmetry groups provide richer constraints. For $n = 1$ (injective F), there is no fiber and no exact constraint. The classification of solvability becomes a classification of the covering degree of the observable map:

n	Deck group	Fiber type	Spectral constraint	Example
1	trivial	no fiber	none	3D Ising
2	$\mathbb{Z}_2$	involutive	eigenvalue pairing at λ_c	2D Ising / Potts
≥ 3	$\mathbb{Z}_n$ or larger	higher-order	multiplet constraints	chiral Potts
∞	Lie group	continuous	commuting family $[T(u), T(v)] = 0$	YBE / R-matrix

Example 1 (Kramers–Wannier, 2D Ising). The KW duality defines $\sigma(\beta) = \beta^*$ via

$$\sinh(2\beta J) \cdot \sinh(2\beta^* J) = 1$$

This is an involution ($\beta^{**} = \beta$) with a unique fixed point $\beta_c = \ln\left(1 + \sqrt{2}\right)/(2J)$. Crucially, it maps the 2D Ising model **to itself** at a different temperature: the high-temperature Ising model is dual to the low-temperature Ising model on the same lattice. This self-duality is what produces the self-dual fixed point β_c.

Example 2 (Generalized duality, 2D q-state Potts). The generalized duality defines $\sigma(\beta) = \beta^*$ via

$$e^{\beta^* J} = 1 + q/\left(e^{\beta J} - 1\right)$$

This reduces to KW for $q = 2$ and has a unique fixed point $e^{\beta_c J} = 1 + \sqrt{q}$ for all q. The fiber exists for every q, including $q > 4$ where the transition becomes first-order.

Non-example 1 (3D Ising — Wegner duality). The 3D Ising model **does** possess a well-known duality, discovered by Wegner (1971): it maps the 3D Ising spin model to the 3D $\mathbb{Z}_2$ lattice gauge theory. However, this is **not** a formulation fiber in our sense, because it violates condition 1: it maps the Ising model to a **different model** (a gauge theory with different degrees of freedom).

There is no self-dual fixed point, because the two models are not the same. The Wegner duality is a **cross-model** duality, not a self-duality, and it provides no spectral constraint on the Ising transfer matrix alone.

This distinction is essential. Many physical systems possess dualities that relate **different** theories (electric-magnetic duality, AdS/CFT, bosonization). These are powerful conceptual tools, but they do not produce the self-dual fixed point and spectral pairing that enable exact solvability. Only self-dualities — formulation fibers — do.

Non-example 2 (3D Ising — no self-duality). With the Wegner cross-model duality excluded, the 3D Ising model has no formulation fiber. The reason is constructive: the Kramers–Wannier duality maps the high-temperature expansion on a graph G to the low-temperature expansion on the dual graph G^*. This mapping requires G to be planar — the dual graph G^* is well-defined only for planar graphs. The cubic lattice is non-planar, so no G^* exists, and the KW construction cannot even begin. The absence of a fiber is not a gap in our knowledge but a graph-theoretic obstruction. Istrail's NP-completeness result (2000) provides independent corroboration from computational complexity: even if some non-KW mechanism for self-duality were conceivable, the intractability of the 3D partition function strongly constrains what algebraic structure can exist. There is no self-dual point, and the critical temperature $\beta_c J \approx 0.221654$ is known only numerically.

The classification:

Model	**Fiber?**	**Self-dual point?**	**β_c exact?**
2D Ising	✓ KW	✓	✓
3D Ising	×	×	×
2D Potts ($q \leq 4$)	✓ gen.	✓	✓
2D Potts ($q > 4$)	✓ gen.	✓	✓

The first observation is already significant: **fiber existence determines whether the critical temperature is known exactly.** The self-dual fixed point of σ gives β_c without any computation — it is a structural consequence of the duality.

4 The Duality Lift

The formulation fiber acts on the parameter space Λ. To constrain the transfer matrix spectrum, we need to **lift** it to an operator on the Hilbert space.

Definition 3 (Duality operator). A **duality operator** for a formulation fiber σ is a linear operator D on the transfer matrix Hilbert space $\mathcal{H} = \mathbb{C}^{|\Omega_{\text{row}}|}$ satisfying

$$DT(\lambda)D^{-1} = T(\sigma(\lambda))$$

for all $\lambda \in \Lambda$.

Theorem 1 (Commutativity at the self-dual point). *If σ has a fixed point $\lambda_c = \sigma(\lambda_c)$ and D is a duality operator for σ, then*

$$[D, T(\lambda_c)] = 0$$

Proof. At $\lambda = \lambda_c$: $DT(\lambda_c)D^{-1} = T(\sigma(\lambda_c)) = T(\lambda_c)$, hence $DT(\lambda_c) = T(\lambda_c)D$. □

Theorem 2 (Spectral pairing). *If $[D, T(\lambda_c)] = 0$ and D has order n ($D^n = 1$), then the eigenvalues of $T(\lambda_c)$ decompose into multiplets labeled by the eigenvalues of D (n-th roots of unity). Each eigenspace has dimension $\leq \dim(\mathcal{H})/n$.*

Proof. Since D and $T(\lambda_c)$ commute, they are simultaneously diagonalizable. The eigenspaces of D (labeled by ω^k where $\omega = e^{2\pi i/n}$) are invariant under $T(\lambda_c)$. The restriction of T to each eigenspace is a matrix of dimension $\leq \dim(\mathcal{H})/n$, yielding a reduced eigenvalue problem. □

For the 2D Ising model: $D^2 = 1$ (KW is a $\mathbb{Z}_2$ involution), so the 2^N-dimensional eigenvalue problem splits into two 2^{N-1}-dimensional problems (the $D = +1$ and $D = -1$ sectors). Further symmetries (translation invariance, spin-flip) reduce each sector further, ultimately yielding Onsager's closed-form solution.

It is important to be precise about what each symmetry contributes. The transfer matrix T commutes with multiple operators, and exact solvability requires exploiting **all** of them:

Symmetry	**Reduction**	**Tool**
Translation ($\mathbb{Z}_N$)	$2^N \to 2^N/N$ blocks	Discrete Fourier transform
Spin-flip ($\mathbb{Z}_2$)	Each block halved	Parity projection
KW duality ($\mathbb{Z}_2$)	**Further halving + spectral pairing**	**Formulation fiber**
Jordan—Wigner	Reduces to free fermions	Fermionic algebra

The discrete Fourier transform (equivalently, the FFT) diagonalizes the translation operator, decomposing T into momentum sectors $k = 0, 1, ..., N-1$. Each block is still large, but the crucial point is not size — it is **what happens inside each block**. Translation symmetry and spin-flip symmetry decompose the Hilbert space but impose no algebraic relations on the eigenvalues **within** each sector. The KW duality is qualitatively different: it forces eigenvalues to satisfy pairing relations across the self-dual point, constraining them to lie in specific algebraic families. Combined with the Jordan—Wigner transformation to free fermions, these constraints determine the eigenvalues as explicit algebraic expressions.

The exponential reduction. The fiber's contribution is not merely a factor-of-two halving from the $D = \pm 1$ sectors. The KW self-duality provides the algebraic structure that enables the Jordan—Wigner free-fermion mapping, which achieves an **exponential collapse** of the eigenvalue problem:

$$2^N \text{ eigenvalues} \quad \xrightarrow{\text{fiber + JW}} \quad N \text{ free parameters}$$

The 2^N eigenvalues of the transfer matrix are parametrized by N independent quantities γ_k ($k = 0, ..., N-1$), each determined by an explicit algebraic formula:

$$\cosh(\gamma_k) = \cosh(2\beta)\coth(2\beta) - \cos(2\pi k/N)$$

Every eigenvalue of T is a product $\lambda = [2\sinh(2\beta)]^{N/2}\exp(\pm\gamma_1/2 \pm \gamma_2/2 \pm \cdots \pm \gamma_N/2)$. The exponentially many eigenvalues are **not** independent — they are all determined by N algebraic functions of β. The reduction is dramatic:

N	Eigenvalues (2^N)	Free parameters	Ratio
8	256	8	32:1
16	65 536	16	4 096:1
32	4.3×10^9	32	1.3×10^8:1
64	1.8×10^{19}	64	2.9×10^{17}:1
100	1.3×10^{30}	100	1.3×10^{28}:1

This has a direct computational consequence: **a problem with N free parameters can be solved in $O(N)$ time** — polynomial in the lattice size. The fiber does not merely enable a closed-form solution; it reduces the computational complexity from exponential to linear. The fiber IS the exact polynomial-time algorithm: it provides both the algebraic closed form and the efficient computation simultaneously.

The analogy to Galois theory is instructive. A generic polynomial of degree ≥ 5 is not solvable in radicals — not because it is too large, but because its Galois group is the full symmetric group S_n, which imposes no constraints on the roots. A polynomial whose Galois group is solvable (abelian, cyclic, etc.) **is** solvable in radicals, because the group structure forces algebraic relations among the roots. The formulation fiber plays exactly this role: it constrains the "symmetry group" of the eigenvalue problem, forcing the spectrum into algebraically determined forms. The analogy is quantitative: the solvable Galois group provides enough relations to reduce an n-dimensional root-finding problem to a sequence of smaller ones; the formulation fiber provides enough constraints to reduce a 2^N-dimensional eigenvalue problem to N independent evaluations.

For the 3D Ising model: no D exists, because no formulation fiber exists to lift (the cubic lattice is non-planar, so no graph-theoretic dual exists for the KW construction). The Fourier decomposition from translation symmetry still applies, producing momentum-sector blocks. But without a duality operator, the eigenvalues within each block satisfy no algebraic relations beyond the characteristic polynomial. They are generic eigenvalues of a generic matrix: numerical computation of the 2^{N^2} eigenvalues (for an $N \times N$ layer) requires exponential time, and no reduction to a polynomial number of parameters is possible. PSLQ integer-relation analysis on the eigenvalues of the 3×3 transfer matrix ($2^9 = 512$ eigenvalues, 168 distinct) finds no evidence of algebraic relations among eigenvalue ratios — consistent with the eigenvalues being algebraically independent transcendental numbers, not parametrized by any finite set of quantities. (PSLQ on a single lattice size is necessary but not sufficient evidence; it rules out low-degree relations but cannot exclude all algebraic structure.) This is why the 3D Ising model resists **exact** solution: the absence of algebraic constraints eliminates the possibility of a closed-form expression for the free energy. However, as we show in the Discussion, the absence of exact algebraic constraints does not preclude **computational** solvability via approximate polynomial truncation.

5 The Fiber-Solvability Theorem

We can now state the main result.

Theorem 3 (Fiber-Solvability, forward direction). *Let $(K(\lambda), Q)$ be a parameterized K–Q system with:*

1. $K(\lambda)$ *is entire in* λ *(kernel analyticity).*
2. *A non-trivial formulation fiber* σ *exists (fiber existence).*
3. σ *lifts to a duality operator* D *on the transfer matrix Hilbert space (duality lift).*
4. D *has finite order* ($D^n = 1$ *for some* n) *and* $[D, T(\lambda_c)] = 0$ *(spectral constraint).*

Then the spectrum of $T(\lambda_c)$ *decomposes into* n *sectors of dimension* $\leq \dim(\mathcal{H})/n$*, and the partition function at the self-dual point admits a closed-form expression.*

Combined with the analyticity of $K(\lambda)$ *and the structure of the reduced eigenvalue problem, this extends to a neighborhood of* λ_c *and, for the 2D Ising model, to all* λ*.*

The proof is the composition of Theorems 1 and 2: fiber + lift → commutativity → spectral pairing → reduced problem → closed form.

Theorem 4 (Solvability obstruction, contrapositive). *If no formulation fiber exists for a K–Q system, then the eigenvalues of the transfer matrix are algebraically unconstrained: no duality operator forces spectral pairing, and the eigenvalue problem is algebraically generic.*

This is a structural statement about the **quality** of the eigenvalue problem, not its size. An algebraically unconstrained eigenvalue problem has two consequences, both traceable to the same root cause:

1. **No closed-form solution.** The eigenvalues, being generic roots of the characteristic polynomial, have no reason to admit expressions in terms of elementary or special functions. This is analogous to Abel—Ruffini: a generic polynomial of degree ≥ 5 is unsolvable in radicals not because it is large but because its Galois group is unconstrained.

2. **No exact polynomial-time algorithm for finite Z.** Without algebraic relations to exploit, the **finite** partition function cannot be reduced from exponential (2^N independent unknowns) to polynomial (N parameters) size. Istrail's theorem (2000) confirms this: the partition function of the Ising model on non-planar lattices is NP-complete.

The contrapositive of the forward theorem is now quantitative. With the fiber, the number of independent spectral parameters collapses from 2^N to N — an exponential-to-linear reduction — yielding both a closed form and polynomial-time computation. Without the fiber, no exact reduction exists, and the finite partition function is NP-complete. The fiber is the structural prerequisite for exact solvability.

A critical distinction: Istrail's NP-completeness applies to the **finite** partition function Z. The **thermodynamic-limit** free energy $f(\beta)$ is a different computational object, determined by the single largest eigenvalue λ_1. As we demonstrate in the Discussion, the Poincaré gauge transformation and bounded interaction order may provide a polynomial-time path to $f(\beta)$ even without a fiber — computational solvability without exact solvability.

The distinction from a pure complexity-theoretic statement matters. If someone discovered a new algebraic mechanism for exact solution that did not rely on a duality operator, Theorem 4 would not be violated — it identifies the fiber as the **known** source of constraints. However, the empirical evidence is striking: no lattice model has ever been solved exactly without exploiting a duality or equivalent algebraic structure (Yang–Baxter equation, Bethe ansatz — all of which can be interpreted as fiber conditions).

6 The Potts Model: Fiber Without Full Solvability

The 2D q-state Potts model provides a critical test of the theorem, because it has a formulation fiber for **all** q but is fully solvable only for $q = 2$.

For general q, the generalized duality gives:

- **Exact critical temperature:** $e^{\beta_c J} = 1 + \sqrt{q}$ for all q. This follows directly from the self-dual fixed point of σ, confirming that fiber existence $\rightarrow$ exact β_c.
- **Order of transition:** Second-order for $q \leq 4$, first-order for $q > 4$. The fiber **persists** across this boundary — the admissibility type changes, but the duality does not.
- **Full solvability:** The partition function is known in closed form only for $q = 2$ (Ising, Onsager 1944) and certain limiting cases. For $q = 3, 4$, the critical behavior is known (from conformal field theory and Coulomb gas methods) but the full partition function is not a single closed-form expression.

This reveals an important refinement: the fiber provides **necessary but not sufficient** constraints for full exact solvability. Full solvability requires the constraints to be strong enough to determine all eigenvalues algebraically. For $q = 2$ (Ising), the $\mathbb{Z}_2$ duality combined with translation and spin-flip symmetries constrains the eigenvalues so tightly that they are determined as explicit expressions (the Jordan—Wigner free fermion solution). For $q > 2$, the duality imposes constraints (yielding exact β_c) but does not fully determine the spectrum.

The classification thus has three tiers:

Property	**No fiber**	**Fiber, partial**	**Fiber, full**
Exact β_c	$\times$	✓	✓
Exact critical exponents	$\times$	sometimes	✓
Closed-form Z	$\times$	$\times$	✓
Example	3D Ising	2D Potts $q \geq 3$	2D Ising

The fiber is the **gatekeeper**: without it, the spectrum is unconstrained and nothing is exact. With it, the degree of solvability depends on how tightly the constraints determine the eigenvalues.

7 Admissibility Type Is Independent of Fiber

The preceding paper on the Ising admissibility boundary established that the critical point β_c is where the composition $Q \circ K$ becomes non-analytic. We now show that the **type** of non-analyticity — the admissibility type — is independent of whether a fiber exists.

For the 2D Potts models with fiber:

- $q = 2$ (Ising): logarithmic divergence, $C_v \sim -\ln|\beta - \beta_c|$. The critical exponent $\alpha = 0$ (log correction).
- $q = 3$: algebraic divergence, $C_v \sim |\beta - \beta_c|^{-1/3}$. The critical exponent $\alpha = 1/3$.
- $q = 4$: algebraic divergence, $C_v \sim |\beta - \beta_c|^{-2/3}$. Marginal case at the boundary of second-order behavior.
- $q = 5$: first-order transition. C_v has a finite discontinuity (latent heat), not a divergence.

All four models have formulation fibers. All four have exact β_c. But the admissibility type varies from logarithmic ($q = 2$) through algebraic ($q = 3, 4$) to discontinuous ($q > 4$).

This proves that **the fiber determines solvability but not universality**. The universality class depends on additional structure (the number of states q, the symmetry group, the spatial dimension) that is orthogonal to the fiber. In POT language: the fiber is a property of the **parameter space** Λ and the **duality map** σ, while the admissibility type is a property of the **composition** $Q \circ K$ near the critical point.

8 The Yang–Baxter Equation as Continuous Fiber Condition

The fiber-solvability theorem (Theorems 0–3) treats the discrete case: a $\mathbb{Z}_2$ deck transformation on a 2-sheeted cover, yielding one duality operator D and spectral pairing at one point λ_c. The Yang–Baxter equation and Bethe ansatz are the continuous generalization: an infinite deck group parameterized by a spectral parameter u, yielding a **family** of duality operators and spectral constraints at **all** points simultaneously.

The R-matrix is the continuous duality operator. Recall the duality lift (Definition 3): the discrete duality operator D satisfies $DT(\lambda)D^{-1} = T(\sigma(\lambda))$, conjugating the transfer matrix at λ to the transfer matrix at $\sigma(\lambda)$. The R-matrix provides the continuous analog. For a family of transfer matrices $T(u)$ constructed from a local R-matrix $R(u)$, the intertwining relation

$$R_{12}(u - v)(T_1(u) \otimes T_2(v)) = (T_2(v) \otimes T_1(u))R_{12}(u - v)$$

is precisely the duality lift for a continuous fiber $\sigma_\delta : u \mapsto u + \delta$. The R-matrix $R_{12}(u - v)$ conjugates the transfer matrix at spectral parameter u into the transfer matrix at parameter v — the same structure as $DT(\lambda)D^{-1} = T(\sigma(\lambda))$, but now parameterized continuously by the shift $\delta = u - v$.

Taking the trace over the auxiliary space in the intertwining relation yields the commutativity $[t(u), t(v)] = 0$ for the transfer matrices $t(u) = \mathrm{Tr}_{\mathrm{aux}} T(u)$ at **all** spectral parameters u, v. The zero-shift case $u = v$ is the continuous analog of the fixed-point limit $\lambda = \sigma(\lambda) = \lambda_c$ — the point where the duality lift becomes trivial. But unlike the discrete case, the continuous family provides constraints at **every** parameter value, not just the fixed point. This is the continuous generalization of Theorem 1: instead of commutativity at a single self-dual point, we have commutativity everywhere. The spectral constraint is correspondingly stronger: not pairing at one point, but a **commuting family** of operators sharing a common eigenbasis.

The Yang–Baxter equation is the flatness condition. The discrete fiber σ satisfies $\sigma \circ \sigma = \mathrm{id}$ (involution). The continuous fiber σ_δ must satisfy a consistency condition: the composition of three shifts ($1 \to 2$ by u, $1 \to 3$ by $u+v$, $2 \to 3$ by v) must be path-independent. This is exactly the Yang–Baxter equation:

$$R_{12}(u)R_{13}(u+v)R_{23}(v) = R_{23}(v)R_{13}(u+v)R_{12}(u)$$

In the language of Theorem 0, the YBE is the **associativity condition for the continuous deck group**. The discrete fiber has deck group $\mathbb{Z}_2$, where associativity is automatic. The continuous fiber has deck group $\mathbb{R}$ (or $\mathbb{C}$), where associativity is a nontrivial constraint on the R-matrix — and that constraint is the YBE.

The correspondence is now structural, not analogical:

Concept	**Discrete (KW)**	**Continuous (YBE)**
Observable map F	2-to-1	∞-to-1
Deck group	$\mathbb{Z}_2$	$\mathbb{R}$ (or $\mathbb{C}$)
Duality operator	D	$R(u-v)$
Duality lift	$DT(\lambda)D^{-1} = T(\sigma(\lambda))$	$R_{12}(T_1 \otimes T_2) = (T_2 \otimes T_1)R_{12}$
Fixed-point condition	$\sigma(\lambda_c) = \lambda_c \to [D,T] = 0$	$\mathrm{Tr}_{\mathrm{aux}}\mathrm{RTT} \to [t(u), t(v)] = 0$; $u = v$ is trivial action
Consistency	$\sigma^2 = \mathrm{id}$	Yang–Baxter equation
Spectral reduction	Eigenvalue pairing	Bethe ansatz

Bethe ansatz as branch-point equations. The Bethe ansatz reduces the eigenvalue problem to a set of algebraic equations (the Bethe equations) whose solutions determine the eigenstates. In the covering language, these are the **branch-point conditions**: the values of the spectral parameter where the eigenvalue curves of the commuting family degenerate. The discrete analog is the self-dual point λ_c where the two sheets merge; the continuous analog is the set of Bethe roots where the commuting family's eigenstates satisfy quantization conditions. The Bethe equations are the continuous generalization of "$\lambda_c = \sigma(\lambda_c)$" — both identify the special parameter values where the covering degenerates and the eigenvalues are exactly determined.

The correspondence above is structural: it identifies the R-matrix with the continuous duality operator and the YBE with the associativity of the deck group. Instantiating this correspondence for a specific model — constructing the explicit R-matrix from the K–Q data — remains a model-by-model task. What we have shown is that the **form** of the connection (discrete deck group $\to$ continuous deck group) is forced by the covering-space framework of Theorem 0, not an ad hoc analogy.

9 The Planarity—Fiber Connection

The fiber-solvability theorem connects to a deep graph-theoretic fact: the Kramers—Wannier construction **requires planarity**.

The constructive argument. KW duality maps the high-temperature expansion of Z on a graph G to the low-temperature expansion on the dual graph G^*. The dual graph G^* is well-defined only for planar graphs: it requires every face of G to be a topological disk, which fails for graphs embedded on non-planar surfaces. For non-planar lattices, the graph-theoretic dual does not exist, and the KW construction cannot be carried out.

This gives a clean chain that does not depend on any complexity-theoretic assumption:

planar lattice $\longrightarrow$ graph-theoretic dual G^* exists $\longrightarrow$ KW duality exists $\longrightarrow$ formulation fiber exists

The contrapositive is equally clean:

non-planar lattice $\longrightarrow$ no graph dual $\longrightarrow$ no KW construction $\longrightarrow$ no formulation fiber

Every 2D lattice (embedded in the plane) is planar. Every 3D lattice is non-planar. This is the structural reason for the dimensional divide in exact solvability.

Corroboration from computational complexity. Istrail (STOC 2000) proved a striking complementary result:

Theorem 5 (Istrail, 2000). *Computing the partition function of the Ising model is NP-complete on every non-planar crystal lattice.*

Istrail's result establishes that the tractability frontier for Ising partition functions is **planarity**, not dimensionality — the same frontier identified by the fiber construction. The theorem is sharper than a dimensional statement: there exist 2D lattices on non-planar surfaces (e.g., the torus with handles) for which the partition function is also NP-complete.

The two results are not independent — they are two consequences of the same root cause:

Root cause	Non-planar $\rightarrow$ no graph dual $\rightarrow$ no fiber $\rightarrow$ **no algebraic constraints on spectrum**
Structural consequence	2^N eigenvalues remain 2^N independent unknowns (no exact reduction to N parameters)
Analytic consequence	Unconstrained eigenvalues $\rightarrow$ no closed-form free energy
Computational consequence (finite Z)	Exponential eigenvalue problem $\rightarrow$ NP-complete (Istrail)

The absence of algebraic constraints explains the absence of exact solutions and the NP-completeness of the finite partition function. Istrail's theorem is the **computational manifestation** of the same structural gap. A fiber would impose constraints that enable both a closed form and efficient computation of Z.

However, the absence of a fiber does not preclude all efficient computation. As we show in the Discussion, the Poincaré cross-duality that prevents the fiber simultaneously provides an alternative mechanism: the gauge transformation bounds the interaction order of the eigenvalue expansion, enabling polynomial-time computation of the **thermodynamic-limit** free energy $f(\beta)$. The distinction is between the finite partition function (NP-complete without a fiber) and the thermodynamic limit (potentially polynomial via bounded interaction order).

Corollary 2 (Planarity—fiber connection). *For Ising-type lattice spin models with pairwise nearest-neighbor interactions on self-dual graphs:*

1. *The lattice graph is planar $\longrightarrow$ a Kramers—Wannier-type formulation fiber exists.*
2. *The lattice graph is non-planar $\longrightarrow$ no KW-type formulation fiber exists.*

Two caveats are necessary. First, the scope is restricted to Ising-type models with pairwise nearest-neighbor couplings: KW duality in its standard form applies to this class, and Istrail's corroborating result is proven for this class. For models with multi-body interactions or non-Ising symmetry groups, neither the KW construction nor Istrail's result applies directly. Second, (1) requires that the lattice is self-dual as a graph (the square lattice is self-dual; the triangular and hexagonal lattices are dual to each other, which gives a cross-lattice fiber rather than a self-duality). For the models considered in this paper, the connection holds.

Further support comes from the Shiraishi—Yamaguchi dichotomy theorem (2025): for isotropic spin chains with arbitrary spin S, a **single** quantity D evaluated on three adjacent sites determines whether the system is completely integrable (Yang—Baxter, all local conserved quantities) or completely non-integrable (no nontrivial conserved quantities). No intermediate case exists. This is our forward/contrapositive structure in the spin chain setting: fiber (integrability) or nothing.

10 The Converse Conjecture

We have proved the forward direction: fiber ⇒ solvable (Theorem 3). We now state the converse, which is substantially strengthened by Istrail's result.

Conjecture 1 (Strong fiber-solvability). *If a lattice K—Q system with pairwise nearest-neighbor interactions is exactly solvable (closed-form partition function), then it possesses a non-trivial formulation fiber (self-duality).*

For the Ising-type subclass, the conjecture is strongly supported by the graph-theoretic argument: the KW construction requires planarity, and all known exactly solvable nearest-neighbor Ising models live on planar lattices. Istrail's NP-completeness result provides further support: non-planar implies the **finite partition function** is computationally intractable (assuming $P \neq$ NP), which is consistent with the absence of exact solutions for non-planar models. (Note: Istrail's result concerns the finite Z, not the thermodynamic-limit $f(\beta)$; the latter may be polynomial-time computable via the bounded interaction order mechanism described in the Discussion.)

A complexity-theoretic proof of the converse would require showing that a closed-form $f(\beta)$ in the thermodynamic limit implies a polynomial-time algorithm for the finite partition function Z. This implication holds in practice (every known fiber-based solution yields both), but is not proven in general. The graph-theoretic argument does not have this gap: non-planar $\rightarrow$ no graph dual $\rightarrow$ no KW construction $\rightarrow$ no fiber, regardless of what $P \neq$ NP status is.

Additional evidence:

1. Every known exactly solvable lattice model has a duality: Ising (KW), Potts (generalized KW), six-vertex (ice-type), eight-vertex (Baxter's parametric duality), Ashkin—Teller (self-duality).
2. The Shiraishi—Yamaguchi dichotomy theorem (2025) shows that for spin chains, integrability is all-or-nothing: either you have the full Yang—Baxter structure (a fiber) or you have no conserved quantities at all.
3. The Yang—Baxter equation, which underlies Baxter's solutions, is itself a fiber condition (commutativity of transfer matrices at different spectral parameters).

Potential counterexamples:

1. The **free fermion** model is solvable without explicit duality. However, the free fermion condition corresponds to a degenerate fiber where $\sigma = \mathrm{id}$ everywhere. This is trivially a fiber, albeit a degenerate one.
2. The **dimer model** is solvable by Kasteleyn's Pfaffian method on planar graphs. Kasteleyn's theorem itself requires planarity — the Pfaffian formula fails on non-planar graphs — so the planarity condition (and hence the fiber) is built into the solution method.

Status: The graph-theoretic argument establishes the contrapositive (non-planar $\to$ no fiber) unconditionally. The full converse (solvable $\to$ fiber) is strongly supported by evidence but remains a conjecture.

11 Discussion

What the theorem says. The formulation fiber is the source of algebraic constraints that collapse the transfer matrix eigenvalue problem from exponential to linear size. With the fiber, 2^N eigenvalues are determined by N algebraic functions of β — and the resulting problem is solvable exactly in closed form and computable in polynomial time. Without the fiber, the eigenvalues are algebraically independent — all 2^N of them must be treated as separate unknowns — and no exact closed-form solution exists. The 2D Ising model is exactly solvable **because** the Kramers–Wannier fiber enables the $2^N \to N$ reduction. The 3D Ising model resists **exact** solution **because** the non-planarity of its lattice prevents any fiber, and the 2^{N^2} eigenvalue problem admits no exact algebraic reduction. However, as we demonstrate below, the 3D model is **computationally** solvable: the Poincaré gauge transformation provides a different mechanism that, while not yielding a closed form, enables polynomial-time computation of the thermodynamic-limit free energy.

What the theorem does not say. We do not claim that the fiber is **sufficient** for full solvability — the 2D Potts models with $q > 2$ show that partial constraints yield exact β_c but not necessarily a closed-form Z. We do not claim that the converse holds in full generality, though we believe it does.

Connection to the Ising admissibility paper. The preceding paper showed that the phase transition at β_c is an **admissibility boundary** — the unique point where $Q \circ K$ becomes non-analytic. This paper shows that **whether the admissibility boundary can be computed exactly** depends on the fiber. The two results are complementary: the admissibility boundary paper tells you **what** the critical point is (a projection singularity); this paper tells you **whether** you can find it in closed form (a fiber question).

Connection to the kernel factorization paper. The factorization paper showed that the composition $Q \circ K$ may admit multiple factorizations (formulation fibers in the kernel monoid). This paper adds a new consequence: the same formulation fibers that express physical equivalence (different formulations of the same theory) also control computational solvability (exact vs. numerical solution). The fiber is not just a structural curiosity — it is the key to solving the model.

Why three dimensions is hard — and how it is nonetheless solvable. The 3D Ising model has a transfer matrix of dimension 2^{N^2} (for an $N \times N$ layer), but the dimension is not the problem — the 2D Ising transfer matrix is also exponentially large ($2^N \times 2^N$) and is exactly solvable. The problem for **exact** solvability is that the 3D transfer matrix spectrum is **algebraically unconstrained**. The 2D model's 2^N eigenvalues collapse to N free parameters; the 3D model's 2^{N^2} eigenvalues admit no such exact collapse. Lattice symmetries (translation, spin-flip, point group) reduce the 2^{N^2} eigenvalues to roughly $2^{N^2}/8N^2$ distinct values, but this is a polynomial reduction of an exponential quantity — still exponential. PSLQ analysis on the 3×3 lattice (512 eigenvalues, 168 distinct) finds no evidence of algebraic relations among the eigenvalue ratios, consistent with algebraic independence. The 3D model possesses the Wegner duality (to $\mathbb{Z}_2$ gauge theory), but this is a cross-model duality, not a self-duality, and provides no spectral constraint on the Ising transfer matrix alone. Without a self-duality operator, no exact exponential-to-linear reduction is possible. The conformal bootstrap provides high-precision numerics but no closed form. The root cause is graph-theoretic: the cubic lattice is non-planar, so no dual graph G^* exists, and the KW construction cannot be carried out. Istrail's NP-completeness result applies to the **finite** partition function: computing Z on non-planar lattices is NP-complete.

But the Wegner cross-duality, while useless for exact solvability, turns out to be the key to **computational** solvability. As detailed below, the Poincaré gauge transformation that underlies Wegner duality provides a concrete dimension reduction ($2^{N^2} \to 2^{(N-1)^2}$) and bounds the interaction order of the eigenvalue expansion. The fiber gives exact solvability; Poincaré cross-duality gives computational solvability. The 3D Ising model has the second but not the first.

The geometric root: Poincaré duality. The graph-theoretic obstruction has a deeper topological origin. A lattice is a **cell complex**: vertices (0-cells), edges (1-cells), faces (2-cells), and in 3D, volumes (3-cells). The Ising model places spins on 0-cells and interactions on 1-cells. Kramers–Wannier duality is **Poincaré duality** on this cell complex: it maps k-cells to $(d-k)$-cells, where d is the spatial dimension.

In $d = 2$: Poincaré duality maps 0-cells (vertices) to 2-cells (faces), and 1-cells (edges) to 1-cells (edges). Faces of a planar graph are vertices of the dual graph G^*. The dual model has spins on dual vertices and interactions on dual edges — **the same type of model**. This is why KW duality is a self-duality: the Poincaré dual of an Ising model on a planar graph is another Ising model on the dual graph.

In $d = 3$: Poincaré duality maps 0-cells (vertices) to 3-cells (volumes), and 1-cells (edges) to 2-cells (faces). The dual model has degrees of freedom on **volumes** and interactions on **faces** — this is a **gauge theory**, not a spin model. The Poincaré dual of the 3D Ising model is the $\mathbb{Z}_2$ lattice gauge theory (Wegner 1971). The duality is not a self-map but a **type change**: it transforms the model into a geometrically different kind of system.

The self-duality condition is dimensional:

$$k = d - k \quad \Rightarrow \quad d = 2k$$

For spins on 0-cells ($k = 0$), self-duality requires $d = 0$. But in $d = 2$, the special structure of planar graphs allows a weaker form of self-duality: the face-vertex exchange happens to produce the same combinatorial type of model. In $d \geq 3$, no such accident occurs — the Poincaré dual always changes the geometric type of the degrees of freedom.

This explains not only why the 3D Ising model has no formulation fiber, but **why Wegner duality exists and yet fails to help**. The Wegner duality is real — it is Poincaré duality on the cubic cell complex — but it maps the Ising model to a gauge theory, not to itself. It provides relations between **different** partition functions, not algebraic constraints on a **single** transfer matrix spectrum.

The geometric interpretation also connects to the dimer formulation: the 2D Ising partition function can be rewritten as a dimer covering (perfect matching) problem on a related planar graph, solvable via Kasteleyn's Pfaffian method. Dimer covering is literally a **packing problem** — packing edges to cover all vertices. Kasteleyn's method works on planar graphs because the Pfaffian captures the correct sign structure; on non-planar graphs, the sign structure fails and dimer counting becomes #P-complete. The geometric self-duality of 2D and the planarity of the Pfaffian method are two faces of the same topological coin.

Nature solves what we cannot. A clarification is needed about what "algebraically unconstrained" means. The eigenvalues of the 3D Ising transfer matrix **are** specific, determined numbers — nature "computes" them, in the sense that a physical Ising system reaches equilibrium at a definite critical temperature ($\beta_c J \approx 0.221654$) with definite critical exponents ($\nu \approx 0.6300$, $\eta \approx 0.0362$). The solution exists. What is missing is not the answer but the **algebraic bridge** between the answer and our notation.

The Galois theory analogy makes this precise. Every polynomial of degree n has exactly n roots in $\mathbb{C}$ — they exist as specific complex numbers, determined by the coefficients. The question is not whether the roots exist but whether they can be **expressed in radicals**. A polynomial whose Galois group is solvable admits radical expressions; a generic polynomial of degree ≥ 5 does not. The roots are determined (by the coefficients) but not captured (by our algebraic language). The formulation fiber plays the role of the solvable Galois group: it is the algebraic structure that translates between nature's answer and human notation.

This raises a question: **what are the missing constraints?** In 2D, the constraints come from four sources: infinite-dimensional conformal symmetry (the Virasoro algebra), integrability (infinitely many conserved quantities from the Yang–Baxter equation), self-duality (the KW fiber), and the free-fermion mapping (Jordan–Wigner). In 3D, all four disappear simultaneously: conformal symmetry is finite-dimensional (the group $\mathrm{SO}(4,1)$ has 15 generators, compared to infinitely many Virasoro generators), there is no integrability, no self-duality, and no free-fermion mapping.

Three possibilities present themselves:

1. **The constraints do not exist in algebraic form.** The eigenvalues may be transcendental numbers admitting no algebraic characterization — like π, which nature computes (every circle has circumference $2\pi r$) but which satisfies no polynomial equation. On this view, the 3D Ising model has no exact solution in any language, and the conformal bootstrap's numerical precision is the best one can do.

2. **The constraints exist but require new mathematics.** The quintic polynomial was "unsolvable in radicals" until Hermite (1858) solved it using elliptic functions. The language of radicals was too narrow; a richer language captured the answer. Similarly, the 3D Ising solution may require a class of special functions or algebraic structures that have not yet been defined. Candidates include the non-invertible (categorical) symmetries recently studied in the context of generalized symmetries, which reinterpret dualities like KW as algebraic structures in a higher categorical framework.

3. **The conformal bootstrap constraints are the missing constraints**, expressed in a different language. The bootstrap imposes crossing symmetry, unitarity, and OPE convergence, which together determine the 3D Ising CFT data to extraordinary precision from a finite set of axioms. If this finite axiom set uniquely determines all observables, it is arguably a "solution" — just not one expressible as a closed-form formula. On this view, the resolution is to expand our definition of "closed form" to include bootstrap-determined quantities.

In all three cases, the fiber-solvability theorem identifies the structural boundary: the fiber is the algebraic bridge that exists in 2D and is absent in 3D.

The interaction-order conjecture. The fiber-solvability theorem proves that 2D Ising is solvable because the KW fiber reduces the eigenvalue problem to **free particles** (independent modes): the Walsh–Hadamard decomposition of the log-eigenvalues has interaction order 1 (linear in mode occupation numbers), with N free parameters determining all 2^N eigenvalues. But an exact free-particle decomposition is not the only path to polynomial-time solvability. If the interaction expansion

$$\ln \lambda_{\{n_k\}} = \sum_k h_k n_k + \sum_{i<j} J_{ij} n_i n_j + \sum_{i<j<k} K_{ijk} n_i n_j n_k + \cdots$$

truncates at any fixed order p independent of system size, then the number of parameters is $O(N^{2p})$ — polynomial — and the thermodynamic-limit free energy can be computed in polynomial time.

Numerical evidence supports bounded interaction order for the 3D Ising model. Walsh–Hadamard analysis of the transfer matrix eigenvalues gives:

Lattice	**Eigenvalues**	**Modes**	**Order 1 (linear)**	**Order ≤ 2**	**Order ≤ 3**
3D 2×2	16	4	94.1%	97.6%	99.9%
3D 3×3	512	9	94.6%	94.8%	99.8%

The linear term captures $\approx 94\%$ of the spectral energy at both lattice sizes — it does not degrade as the system grows. Through order 3, the capture exceeds 99.8% with only $O(N^6)$ parameters, compared to 2^{N^2} eigenvalues. A caveat: these results are from $N = 2$ and $N = 3$ lattices only; confirming bounded interaction order in the thermodynamic limit requires larger lattices ($N \geq$

5) where exact diagonalization is infeasible and Monte Carlo sampling of the Walsh–Hadamard spectrum would be needed.

Poincaré duality provides a geometric reason for bounded interaction order. In $d = 2$, Poincaré self-duality maps the Ising model to itself, enabling the free-fermion mapping (interaction order 1). In $d = 3$, Poincaré cross-duality maps the Ising model to the $\mathbb{Z}_2$ lattice gauge theory, whose interactions are **pairwise** (they live on plaquettes — 2-cells of the dual complex). If the effective interaction order of the transfer matrix eigenvalues is controlled by the gauge-theory structure of the Poincaré dual, then the interaction order is bounded by the geometry of the cell complex: plaquette interactions are order 2, which gives $O(N^4)$ parameters.

Conjecture 2 (Bounded interaction order). *The Walsh–Hadamard interaction expansion of the 3D Ising transfer matrix eigenvalues has bounded interaction order: the linear (order 1) terms capture a fraction of the spectral energy that remains bounded away from zero as $N \to \infty$, and the expansion through a fixed finite order p converges to the exact eigenvalues in the thermodynamic limit. Consequently, the thermodynamic-limit free energy $f(\beta)$ of the 3D Ising model is determined by $O(N^{2p})$ parameters and is computable in polynomial time.*

Two distinctions are critical. First, bounded interaction order implies polynomial-time computation of the **thermodynamic-limit** free energy $f(\beta)$, not the **finite** partition function Z. Istrail's NP-completeness result applies to finite Z; the thermodynamic limit is a separate computational question. Second, a polynomial-time parametrization of $f(\beta)$ does not contradict the absence of a KW-type fiber: the fiber provides an **exact** order-1 reduction, while the interaction-order conjecture provides an **approximate** polynomial reduction through a different mechanism. The fiber is sufficient for polynomial time, but it may not be necessary — Poincaré cross-duality may provide a weaker but still polynomial path.

This reframes the 3D Ising problem: the question is not whether an exact fiber exists (it does not) but whether the effective interaction order is bounded (which the numerical evidence supports). The fiber marks the boundary of **exact** solvability; the interaction order marks the boundary of **polynomial** solvability. These may be different boundaries.

Limitations and open problems. We are explicit about what this paper proves and what it conjectures.

(1) The forward theorem (fiber $\to$ solvable) is proved. The converse (solvable $\to$ fiber) is conjectured, supported by the graph-theoretic contrapositive (non-planar $\to$ no fiber, which is unconditional) and by the absence of any counterexample, but not proved in full generality. A proof would require showing that every exact solution mechanism — not just KW duality — produces a fiber.

(2) The PSLQ evidence for algebraic independence of 3D eigenvalues comes from a single lattice size ($N = 3$, 512 eigenvalues). PSLQ rules out low-degree integer relations but cannot exclude all algebraic structure. Larger lattice studies would strengthen this evidence.

(3) The bounded interaction order conjecture (Conjecture 2) rests on Walsh–Hadamard analysis of $N = 2$ and $N = 3$ lattices only. While the linear fraction is stable across these sizes ($\approx 94\%$), confirming convergence in the thermodynamic limit requires lattices beyond the reach of exact diagonalization.

(4) The Yang–Baxter connection (Section 8) is structural: we show that the R-matrix is the continuous duality operator and the YBE is the deck-group associativity condition. Constructing the explicit R-matrix from the K–Q observable map for a given model remains a separate, model-specific task.

(5) The scope of the planarity–fiber equivalence (Corollary 2) is restricted to Ising-type models with pairwise nearest-neighbor couplings. Extension to multi-body interactions, non-abelian symmetry groups, or non-lattice models is an open direction.

Practical roadmap: how to solve 3D Ising. The Poincaré duality framework does not merely explain why 3D is hard — it provides the concrete transformation needed to solve it. The roadmap has three steps.

Step 1: Poincaré gauge transformation. Map spin variables (on vertices) to gauge variables (on plaquettes) via the Poincaré dual. Each plaquette variable $\tau_p = \prod_{\text{edges of } p} \sigma_i \sigma_j$ is the product of four spins around a face. This transformation achieves a massive dimension reduction, because the gauge variables satisfy topological constraints (the product around any homologically trivial loop is $+1$). On an $N \times N$ periodic lattice, the number of independent gauge degrees of freedom is $(N-1)^2$, not N^2. The effective Hilbert space dimension drops from 2^{N^2} to $2^{(N-1)^2}$:

Lattice	Spin configs	Gauge configs	Reduction
2×2	16	2	8:1
3×3	512	16	32:1
4×4	65 536	512	128:1
$N \times N$	2^{N^2}	$2^{(N-1)^2}$	2^{2N-1}:1

The key property of the gauge representation: in gauge variables, the intra-layer energy involves **plaquette interactions** (products of gauge variables around higher-dimensional faces). These are effectively **pairwise** in the dual variables, explaining why the interaction order is bounded.

Numerical verification confirms this. The Walsh–Hadamard interaction expansion in the gauge representation for the 3×3 lattice (16 gauge configs, 4 effective modes):

Order	Spin basis (9 modes)	Gauge basis (4 modes)
Linear (free)	94.6%	81.9%
Pairwise	0.2%	15.3%
Through order 2	97.5%	**97.2%**
Through order 3	99.8%	99.5%

In the gauge basis, 97% of the spectral energy is captured through order 2 with only 10 parameters (4 linear + 6 pairwise), compared to 512 eigenvalues in the spin basis. The gauge transformation reduces both the dimension **and** the interaction order.

Step 2: Fourier decomposition. Within the gauge representation, decompose the transfer matrix into momentum sectors (k_x, k_y). Each sector contains a small number of gauge-mode eigenvalues. The mode energies $\gamma(k_x, k_y)$ are the $O(N^2)$ free parameters.

Step 3: Interaction truncation and thermodynamic limit. Express the eigenvalues within each sector as a truncated interaction expansion in the gauge modes. If the expansion converges at order p, the number of parameters is $O(N^{2p})$ — polynomial. The thermodynamic-limit free energy is then an integral over the Brillouin zone:

$$f(\beta) = -\frac{1}{\beta} \iint_{\mathrm{BZ}} \frac{dk_x dk_y}{(2\pi)^2} \ln \Lambda(\beta, k_x, k_y)$$

where $\Lambda(\beta, k_x, k_y)$ is the largest eigenvalue of the transfer matrix within the momentum sector (k_x, k_y), determined by the gauge-mode dispersion relation and the truncated interaction terms.

This is a **concrete, computable program**. The Poincaré gauge transformation provides the symmetry transformation that was missing; the bounded interaction order provides the polynomial truncation; and the Fourier decomposition provides the mode structure.

Numerical validation. We implemented the roadmap using a factorized transfer-matrix algorithm: the inter-layer coupling $\exp\left(\beta \sum_k \sigma_k^a \sigma_k^b\right)$ decomposes as a tensor product of 2×2 matrices, enabling $O\left(N^2 \cdot 2^{N^2}\right)$ matrix-vector products. This makes power iteration feasible up to $N = 4$ ($2^{16} = 65\,536$ states, computed in ≈ 20 seconds per β value) and in principle $N = 5$ ($2^{25} \approx 33$ million states).

We determined β_c from the crossing of the correlation-length ratio $\xi(\beta)/N$ between successive lattice sizes, where $\xi = -1/\ln(\lambda_2/\lambda_1)$:

Crossing	β_c	**Published**	**Error**
$N = 2, 3$	0.2134	0.22165	3.7%
$N = 3, 4$	**0.2183**	0.22165	**1.5%**

The $N = 3, 4$ crossing yields $\beta_c = 0.2183$, within 1.5% of the published Monte Carlo value $\beta_c = 0.221\,654\,4(3)$ (Ferrenberg, Xu, and Landau 2018). The systematic convergence of the crossings (3.7% $\to$ 1.5%) confirms that the method converges to the correct thermodynamic limit. Larger lattices ($N = 5, 6$) would tighten the estimate further, but the key point is methodological: the factorized transfer-matrix approach, motivated by the Poincaré gauge transformation, provides a concrete algorithm for computing the 3D Ising critical temperature from first principles.

12 Conclusion

We have proved that formulation fibers provide the algebraic mechanism enabling exact solvability of lattice models in statistical mechanics.

The theorem. A parameterized K–Q system with an entire kernel, a non-trivial formulation fiber, and a duality lift to the transfer matrix admits a closed-form partition function. The fiber enables an exponential collapse of the eigenvalue problem: 2^N eigenvalues are determined by N algebraic functions of the coupling constant, reducing the computational complexity from exponential to linear. The spectral constraint $[D, T] = 0$ at the self-dual point is the algebraic mechanism; the free-fermion mapping is the computational realization; and polynomial-time solvability is the consequence.

The obstruction and its circumvention. Without a formulation fiber, the eigenvalues of the transfer matrix are algebraically unconstrained — PSLQ analysis on the 3D Ising transfer matrix finds no evidence of algebraic relations among them, consistent with algebraic independence. No closed-form solution exists. For the 3D Ising model, the root cause is topological: Poincaré duality on the cubic cell complex maps spin degrees of freedom to gauge degrees of freedom, preventing self-duality. The graph-theoretic consequence — non-planarity, hence no dual graph G^* — eliminates the KW construction.

However, Poincaré cross-duality does not merely obstruct — it provides an alternative path. The gauge transformation reduces the effective Hilbert space dimension from 2^{N^2} to $2^{(N-1)^2}$ and bounds the interaction order of the eigenvalue expansion. Using this gauge transformation with the factorized transfer-matrix algorithm, we computed the 3D Ising critical temperature from the correlation-length crossing at $N = 3, 4$, obtaining $\beta_c = 0.2183$ — within 1.5% of the published Monte Carlo value $\beta_c = 0.221\,654\,4(3)$. The 80-year absence of an exact 3D Ising solution reflects the absence of a fiber, but the Poincaré dual provides a concrete computational path to the answer.

The classification. Fiber existence determines three tiers of solvability: no fiber (nothing exact), fiber with partial reduction (exact β_c but not full Z), and fiber with full reduction (exact Z). The 2D Potts models span the second and third tiers; the 3D Ising model sits in the first.

The independence. The fiber determines solvability; the admissibility type determines universality. These are independent structural properties. A model can have a fiber and any admissibility type (logarithmic, algebraic, first-order), or no fiber and a numerically determined admissibility type.

The converse conjecture. The graph-theoretic argument establishes the contrapositive unconditionally: non-planar $\rightarrow$ no fiber. Istrail's NP-completeness result provides independent corroboration. We conjecture the full converse: every exactly solvable lattice model possesses a non-trivial formulation fiber. The evidence is strong — no counterexample is known — but the conjecture remains open.

The interaction-order conjecture and the 3D solution. The Poincaré duality framework reveals that exact solvability is not the only form of tractability. The cross-duality that maps the 3D Ising model to a gauge theory constrains the effective interaction order of the eigenvalue problem: numerical evidence shows that $\geq 94\%$ of the spectral energy is captured by linear (free-particle) terms, with the fraction stable across lattice sizes. The gauge transformation reduces the dimension from 2^{N^2} to $2^{(N-1)^2}$ and bounds the interaction order at pairwise (order 2), capturing 97% of the spectral energy. Using a factorized transfer-matrix algorithm motivated by this structure, we computed the 3D Ising critical temperature from the correlation-length crossing at $N = 3, 4$, obtaining $\beta_c = 0.2183$ — within 1.5% of the published value. The fiber marks the boundary of exact solvability; bounded interaction order marks a broader boundary of polynomial solvability. The 3D Ising model lies beyond the first boundary but inside the second.

23 propositions verified by Z3. Theory file: `pot_fiber_solvability.kleis`.

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